

tra, Haryana, Punjab, and the evolving market in Uttar Pradesh.

“We were a bootstrap company and faced many challenges. We operated on whatever we earned, covering employee salaries, R&D costs, and all operational expenses. It was definitely not easy, which is why the failure rate for startups is 95%. Despite this, we were determined and rigorously improved our systems. Our focus and ability to learn from our mistakes allowed us to define our boundaries and invest our time, effort, and money in the right direction,” shares Anoop.

The startup claims to manufacture 95% of the components for its drones. While they produce most parts in-house, others are sourced from contract manufacturers. Their automated assembly line can produce 15-20 drones per shift, with a monthly capacity of 300 drones, resulting in an annual production of 4000 drones. If they switch to double shifts, production can double.

The startup offers two drone models, and the whole drone package (drone, extra batteries, insurance, pilot certification, bike box, and some spares, etc) is priced between 800,000 and a million rupees. The primary



Anoop Upadhyay, Director & Co-Founder, IoTechWorld Avigation

distinction between the two models is the battery capacity, with one featuring a 25,200mAh battery and the other a 21,000mAh battery. Additionally, minor differences include the inclusion of a dongle that enables data to be pushed to the cloud and to servers.

The startup has sold approximately 1600 drones, with around 1000 of those sold just last year.

Their sales more than double each year. For instance, in 2021, their revenue was 20 million rupees, which grew exponentially to 100 million, then to 440 million, and is now expected to reach 900 million



Deepak Bhardwaj, Director & Co-Founder, IoTechWorld Avigation

in just four years. This demonstrates significant growth from 20 million to 900 million in a short span.

The team consists of approximately 140 people, with 30 dedicated to R&D, around 30 in customer support, about 20 in sales and marketing, and the remaining involved in development and manufacturing.

“Our two new drone models are set to launch soon. The first model, featuring 95% Indian-made components, will be available by the end of this year or possibly in January or February of next year,” concludes Deepak.

—Nitisha Dubey

2 A MADE-IN-INDIA LOW-CODE PLATFORM FOR EMBEDDED SYSTEMS

Mumbai-based startup Smowcode is building a graphical low-code language that allows bidding farewell to the woes of embedded coding through its vast set of libraries.

Snowcode, or smart flow code, is a visual coding language for ESP32 and MSPM0+ micro-controllers. This low-code language is designed specifically for IoT chip coding (firmware). Founded in 2021 by Adhik Mewada, the platform is currently available in two versions: a non-commercial version for hobbyists, students, and enthusiasts without any coding restrictions or payments, and a

commercial version for industry users with a potential cost.

During the final year of his polytechnic education, Mewada interned with an IoT device manufacturer to develop industry-grade firmware programs and microcontroller-based hardware devices. “While my boss gave me a simple logic diagram, I had to write a thousand lines of code for the same! I began thinking of a way to

simplify my work,” shares Mewada.

That set the genesis for Smowcode, and Mewada started building Smow, a flow-based programming language that allows users to create code flows using Smowcode Studio. The language features two types of nodes—event nodes that trigger events and activity nodes that perform tasks, which then produce outputs in the case of first flows. In parallel flows, Smow allows